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SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

I S. Nithesh (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: S. Nithesh

Level-3

MP

Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.
2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.
3. A smart traffic management system is being designed to control traffic signals across a city based on vehicle density, pedestrian movement, and weather conditions. The system must respond to real-time sensor inputs while also learning traffic patterns over time to reduce congestion. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for minimizing congestion and travel time.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

01. Smart Home automation system.

1. types of Intelligent Agents & Functioning

- Simple Reflex Agent: Acts solely on immediate condition-action rules (IF-THEN)

* Function: If motion is detected in room then 'turn ON light'

- Model-Based Reflex Agent: Keeps an internal state to track unobserved aspects of the environment

* Function: Remembers room is occupied even if motion temporarily stops, preventing lights from turning off mid-use.

- Goal-Based Agent: Selects actions to achieve explicit target states.

* Function: Schedules heating to reach 22°C specifically by 7:00 PM.

- Utility-Based Agent:

Evaluates trade-offs using mathematical utility function to measure state desirability

* Function: Balances comfort against energy costs by raising target temperature slightly during peak electricity pricing.

Q2.

Robot Image Navigation

1. uninformed search strategies

• uniform cost search (UCS):

Expands nodes with the lowest path cost $g(n)$ using a priority queue. Guarantees finding the cheapest path when step costs vary.

• Iterative Deepening Search (IDS): Runs Depth-First search with incrementally increase depth limits ($L=0, 1, 2, \dots$). Guarantees the shortest path for equal step costs.

complexity Analysis

Algorithm	Time complexity	Space	optimality
UCS	$O(b^{l+1} \lceil c^*/\epsilon \rceil)$	$O(b^{l+1} \lceil c^*/\epsilon \rceil)$	• optimal for variable step costs
IDS	$O(b^d)$	$O(bd)$	• optimal for uniform step costs.

• Trade-off: UCS is optimal for varying path costs but uses exponential memory, which can crash low-memory embedded hardware. IDS uses minimal linear memory $O(bd)$ by trading off slight extra computation.

3. Smart traffic management system

1. Types of intelligent Agents & Functioning

- Simple Reflex Agent: changes signal status using current real-time sensor triggers.

- Function: IF lane A vehicles > 10 THEN switch light to GREEN.

- Model-Based Reflex Agent: tracks continuous traffic trends and adjacent intersection states over time

- Function: Estimates incoming traffic from upstream intersection to adjust signal timing before jams occurs.

- Goal-Based Agent: Executes signal cycles aimed at reaching a target condition.

- Function: Target: clear emergency lane traffic within 2 minutes.

- cost:

- Utility-Based Agent: Maximizes a multi-objective utility formulas:

$$\text{utility} = - (\text{wait Time} + \text{Emissions} + \text{Pedestrian Delay})$$